Generative Modelling with Stochastic Differential Equations

Exercises, background

Background

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A Stochastic calculus background

Setting. Throughout, $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\ge 0}, \mathbb{P})$ is a filtered probability space and B is an n-dimensional $(\mathcal{F}_t)$-Brownian motion: B is adapted, $B_0 = 0$, $t \mapsto B_t$ is continuous, and for $0 \le s < t$ the increment $B_t - B_s$ is independent of $\mathcal{F}_s$ with law $\mathcal{N}(0, (t-s)\mathbb{I}_n)$. In particular the coordinates $B^1, \dots, B^n$ are independent one-dimensional Brownian motions and increments over disjoint intervals are independent. For a partition $0 = t_0 < \cdots < t_K$ we write $\Delta_k B = B_{t_{k+1}} - B_{t_k}$, $\Delta t_k = t_{k+1} - t_k$, and $|\pi| = \max_k \Delta t_k$ for its mesh. $|\cdot|$ is the Euclidean norm on vectors and the Frobenius norm on matrices.

Exercise A.1 (Quadratic variation of Brownian motion)**.** Let $t > 0$ and let $(\pi_m)_{m\ge 1}$ be partitions of $[0,t]$ with mesh $|\pi_m| \to 0$. Write $S_m^{ij} = \sum_k \Delta_k B^i \, \Delta_k B^j$, the sum running over the intervals of π_m.

(a) Show that $\mathbb{E}(S_m^{ii} - t)^2 = 2\sum_k \Delta t_k^2 \le 2|\pi_m| t$, so $S_m^{ii} \to t$ in L^2 and in probability.

(b) For $i \ne j$, show that $\mathbb{E}(S_m^{ij})^2 = \sum_k \Delta t_k^2$, so $S_m^{ij} \to 0$ in L^2. (This justifies the rule $\mathrm{d}B_t^i \, \mathrm{d}B_t^j = \delta_{ij}\, \mathrm{d}t$.)

(c) Show that if $\sum_m |\pi_m| < \infty$, then $S_m^{ii} \to t$ almost surely.

Hint: if $Z \sim \mathcal{N}(0,1)$ then $\mathbb{E}Z^4 = 3$.

Exercise A.2 (Itô isometry for simple integrands)**.** Let $0 = t_0 < \cdots < t_K = T$ and $H_s = \sum_{k=0}^{K-1} H_k \mathbf{1}_{(t_k, t_{k+1}]}(s)$, where each H_k is an $\mathcal{F}_{t_k}$-measurable $\mathbb{R}^n$-valued random vector with $\mathbb{E}|H_k|^2 < \infty$. Set $M_t = \int_0^t H_s^\top \, \mathrm{d}B_s = \sum_k H_k^\top (B_{t_{k+1}\wedge t} - B_{t_k \wedge t})$.

(a) Show that $\mathbb{E}M_T = 0$ and $\mathbb{E}M_T^2 = \mathbb{E}\int_0^T |H_s|^2 \mathrm{d}s$ (the Itô isometry).

(b) Show that $(M_t)_{t\in[0,T]}$ is a martingale.

Hint: condition on $\mathcal{F}_{t_k}$ and use that $\Delta_k B$ is independent of $\mathcal{F}_{t_k}$.

Background for the next exercises. The following standard facts may be used without proof.

- *Itô integral.* If H is progressively measurable with values in $\mathbb{R}^{m\times n}$ and $\mathbb{E}\int_0^T |H_s|^2 \mathrm{d}s < \infty$, then $t \mapsto \int_0^t H_s \, \mathrm{d}B_s$ is a continuous martingale on $[0,T]$ with mean zero and $\mathbb{E}|\int_0^t H_s \, \mathrm{d}B_s|^2 = \mathbb{E}\int_0^t |H_s|^2 \mathrm{d}s$.

- *SDEs.* Let $b : [0,\infty) \times \mathbb{R}^n \to \mathbb{R}^n$ and $g : [0,\infty) \times \mathbb{R}^n \to \mathbb{R}^{n\times n}$ be measurable, written $b_t(x)$, $g_t(x)$, and consider

$$\mathrm{d}X_t = b_t(X_t)\,\mathrm{d}t + g_t(X_t)\,\mathrm{d}B_t, \qquad X_0 = \xi \ \ (\mathcal{F}_0\text{-measurable}),$$

  i.e. $X_t = \xi + \int_0^t b_s(X_s)\,\mathrm{d}s + \int_0^t g_s(X_s)\,\mathrm{d}B_s$. If on every $[0,T]$ the coefficients are Lipschitz in x uniformly in t and have linear growth, $|b_t(x)| + |g_t(x)| \le K_T(1+|x|)$, there is a unique continuous adapted solution, and $\mathbb{E}\sup_{t\le T}|X_t|^2 < \infty$ if $\mathbb{E}|\xi|^2 < \infty$. We then call the SDE *well posed*, and write $X^{s,x}$ for the solution started at time s from x.

- *Itô's formula.* For real-valued $\varphi \in C^{1,2}([0,\infty)\times\mathbb{R}^n)$ (C^1 in t, C^2 in x),

$$\mathrm{d}\varphi(t,X_t) = \Big(\partial_t\varphi + \langle b_t, \nabla\varphi\rangle + \tfrac{1}{2}\operatorname{tr}\big(g_t g_t^\top \nabla^2\varphi\big)\Big)(t,X_t)\,\mathrm{d}t + \nabla\varphi(t,X_t)^\top g_t(X_t)\,\mathrm{d}B_t,$$

  with b_t, g_t evaluated at X_t (∇, ∇^2 act on x).

- *Semigroup and generator.* For bounded measurable φ, $P_{s,t}\varphi(x) = \mathbb{E}\,\varphi(X_t^{s,x}) = \int \varphi(y)\, p_{s,t}(x, \mathrm{d}y)$ with $p_{s,t}(x, A) = \mathbb{P}(X_t^{s,x} \in A)$, and the solution is Markov: $\mathbb{E}[\varphi(X_t^{r,x}) \mid \mathcal{F}_s] = P_{s,t}\varphi(X_s^{r,x})$ for $r \le s \le t$. A probability measure ν is mapped to $\nu P_{s,t}$, defined by $\int \varphi \,\mathrm{d}(\nu P_{s,t}) = \int P_{s,t}\varphi \,\mathrm{d}\nu$. The generator is

$$\mathcal{L}_t\varphi(x) = \langle b_t(x), \nabla\varphi(x)\rangle + \tfrac{1}{2}\operatorname{tr}(g_t(x)g_t(x)^\top \nabla^2\varphi(x)),$$

  written $\mathcal{L}$ when b, g do not depend on t.

Exercise A.3 (Chapman–Kolmogorov from the Markov property)**.** Assume the SDE above has Lipschitz coefficients with linear growth, so that it is well posed, and let $P_{s,t}$ be its semigroup. Use the Markov property $\mathbb{E}[\varphi(X_t^{r,x}) \mid \mathcal{F}_s] = P_{s,t}\varphi(X_s^{r,x})$ ($r \le s \le t$, φ bounded measurable).

(a) Show that $P_{r,t}\varphi = P_{r,s}(P_{s,t}\varphi)$ for $r \le s \le t$, and write this identity in terms of the kernels $p_{r,t}$.

(b) Assume that b and g do not depend on t. Show that $P_{s,t} = P_{0,t-s}$ for $s \le t$ (you may use that pathwise uniqueness implies uniqueness in law), and deduce that $P_t := P_{0,t}$ satisfies $P_{t+s} = P_tP_s = P_sP_t$.

(c) Show that if $\mathrm{Law}(X_r) = \nu$ then $\mathrm{Law}(X_t) = \nu P_{r,t} = (\nu P_{r,s})P_{s,t}$.

Hint for (b): $W_u = B_{s+u} - B_s$ is a Brownian motion.

Exercise A.4 (The generator from Itô's formula)**.** Let b, g be continuous on $[0,\infty)\times\mathbb{R}^n$, assume the SDE $\mathrm{d}X_t = b_t(X_t)\mathrm{d}t + g_t(X_t)\mathrm{d}B_t$ is well posed, and let $\mathcal{L}_t$ be its generator, as above.

(a) Let $\varphi \in C_c^2(\mathbb{R}^n)$. Show that $M_t = \varphi(X_t) - \varphi(X_0) - \int_0^t \mathcal{L}_r\varphi(X_r)\,\mathrm{d}r$ is a martingale, and deduce Dynkin's formula: for $s \le t$, $\mathbb{E}\varphi(X_t) - \mathbb{E}\varphi(X_s) = \int_s^t \mathbb{E}\big[\mathcal{L}_r\varphi(X_r)\big]\mathrm{d}r$.

(b) Deduce that $\lim_{h\downarrow 0} h^{-1}\big(P_{t,t+h}\varphi(x) - \varphi(x)\big) = \mathcal{L}_t\varphi(x)$ for all $t \ge 0$ and $x \in \mathbb{R}^n$.

(c) For the OU process $\mathrm{d}X_t = -X_t\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$ with $\mathbb{E}|X_0|^2 < \infty$, apply Itô's formula to $|X_t|^2$ and show that $\mathbb{E}|X_t|^2 = e^{-2t}\mathbb{E}|X_0|^2 + n(1 - e^{-2t})$.

Exercise A.5 (Fokker–Planck by integration by parts)**.** Let X solve a well-posed SDE $\mathrm{d}X_t = b_t(X_t)\mathrm{d}t + g_t(X_t)\mathrm{d}B_t$ with b, g continuous, $b_t \in C^1$ and $g_t \in C^2$ in x, and suppose $\mathrm{Law}(X_t)$ has a density $\rho_t(x)$ with $(t,x) \mapsto \rho_t(x)$ in $C^{1,2}((0,\infty)\times\mathbb{R}^n)$. You may use the *weak Fokker–Planck equation* (a consequence of Dynkin's formula, Exercise A.4): for every $\varphi \in C_c^2(\mathbb{R}^n)$ and $t > 0$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi\,\rho_t\,\mathrm{d}x = \int \mathcal{L}_t\varphi\,\rho_t\,\mathrm{d}x.$$

(a) Derive the Fokker–Planck equation in density form,

$$\partial_t\rho_t = -\nabla\cdot(b_t\rho_t) + \frac{1}{2}\sum_{i,j=1}^{n}\partial_i\partial_j\big((g_tg_t^\top)_{ij}\rho_t\big).$$

(b) For $g_t = \sqrt{2}\,\mathbb{I}_n$ and $\rho_t > 0$, show that $\partial_t\rho_t = \nabla\cdot\big(\rho_t(\nabla\log\rho_t - b_t)\big)$.

(c) For the Langevin diffusion, $b = -\nabla U$ and $g = \sqrt{2}\,\mathbb{I}_n$, with $\mu \propto e^{-U}$ (identified with its density), show that $\partial_t\rho_t = \nabla\cdot\big(\rho_t\nabla\log\frac{\rho_t}{\mu}\big) = \nabla\cdot\big(\mu\nabla\frac{\rho_t}{\mu}\big)$, and deduce that $\rho_t \equiv \mu$ is a stationary solution.

Exercise A.6 (The Ornstein–Uhlenbeck process)**.** Let X solve $\mathrm{d}X_t = -X_t\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$, $X_0 = x \in \mathbb{R}^n$ (unless stated otherwise).

(a) Apply Itô's formula to $e^t X_t$ to show that $X_t = e^{-t}X_0 + \sqrt{2}\int_0^t e^{-(t-s)}\,\mathrm{d}B_s$. Why is this the only solution?

(b) Show that $X_t \sim \mathcal{N}\big(e^{-t}x, (1-e^{-2t})\mathbb{I}_n\big)$, i.e. the transition kernel is $p_t(x,\mathrm{d}y) = (2\pi(1-e^{-2t}))^{-n/2}\exp\big(-|y-e^{-t}x|^2/(2(1-e^{-2t}))\big)\mathrm{d}y$, and that X is a Gaussian process.

(c) Show that $\mathrm{Cov}(X_s, X_t) = (e^{-(t-s)} - e^{-(t+s)})\mathbb{I}_n$ for $0 \le s \le t$.

(d) Let W be a standard n-dimensional Brownian motion and $Y_t = e^{-t}(x + W_{e^{2t}-1})$. Show that Y and X have the same law on $C([0,\infty);\mathbb{R}^n)$.

(e) Show that $\gamma_n = \mathcal{N}(0,\mathbb{I}_n)$ is invariant: if $X_0 \sim \gamma_n$ is independent of B, then $X_t \sim \gamma_n$ for all t. Show that $\mathrm{Law}(X_t) \to \gamma_n$ weakly for every $\mathcal{F}_0$-measurable X_0.

Hint for (b): for deterministic continuous f, $\int_0^t f(s)\,\mathrm{d}B_s$ is the L^2 limit of Gaussian Riemann sums.

Exercise A.7 (KL between a drifted and a driftless diffusion)**.** Let $b, x \in \mathbb{R}^n$, $T > 0$. Let $\mathbb{Q}$ be the law on $C_T = C([0,T];\mathbb{R}^n)$ of $X_t = x + bt + \sqrt{2}B_t$ (the solution of $\mathrm{d}X_t = b\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$, $X_0 = x$), and $\mathbb{P}$ the law of $x + \sqrt{2}B_t$. Recall that $\mathrm{KL}(\mu\|\nu) = \int \log\frac{\mathrm{d}\mu}{\mathrm{d}\nu}\,\mathrm{d}\mu$ if $\mu \ll \nu$ and $+\infty$ otherwise, and *Girsanov's theorem*: if θ is progressively measurable with $\mathbb{E}\exp(\frac{1}{2}\int_0^T|\theta_t|^2\mathrm{d}t) < \infty$ (Novikov's condition), then $\mathcal{E}_T = \exp\big(\int_0^T \theta_t^\top\,\mathrm{d}B_t - \frac{1}{2}\int_0^T|\theta_t|^2\mathrm{d}t\big)$ has mean one, and under $\mathrm{d}\tilde{\mathbb{Q}} = \mathcal{E}_T\,\mathrm{d}\mathbb{P}_\Omega$, where $\mathbb{P}_\Omega$ is the underlying probability measure on Ω, the process $B_t - \int_0^t \theta_s\,\mathrm{d}s$, $t \le T$, is a Brownian motion.

(a) Using Girsanov's theorem, show that $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}(\omega) = \exp\big(\frac{1}{2}\langle b, \omega_T - x\rangle - \frac{1}{4}T|b|^2\big)$.

(b) Deduce that $\mathrm{KL}(\mathbb{Q}\|\mathbb{P}) = \mathrm{KL}(\mathbb{P}\|\mathbb{Q}) = \frac{1}{4}T|b|^2$, and compare with the general bound $\mathrm{KL}(\mathbb{Q}\|\mathbb{P}) \le \mathrm{KL}(\nu^1\|\nu^2) + \frac{1}{4}\mathbb{E}_{\mathbb{Q}}\int_0^T|b_t^1 - b_t^2|^2\mathrm{d}t$ for two SDEs $\mathrm{d}X_t = b_t^i(X_t)\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t$, $X_0 \sim \nu^i$.

(c) Compute the KL divergence between the time-T marginals of $\mathbb{Q}$ and $\mathbb{P}$. Data processing gives $\mathrm{KL}(\text{marginals}) \le \mathrm{KL}(\mathbb{Q}\|\mathbb{P})$: why is it an equality here? (*Chain rule:* for laws on a product space, $\mathrm{KL}(\mu\|\nu) = \mathrm{KL}(\mu_1\|\nu_1) + \int \mathrm{KL}(\mu(\cdot\mid x)\|\nu(\cdot\mid x))\,\mu_1(\mathrm{d}x)$, where μ_1, ν_1 are the first marginals and $\mu(\cdot\mid x), \nu(\cdot\mid x)$ the conditional laws.)